1

I : num1, num2
P : product = num1 * num2
O : product

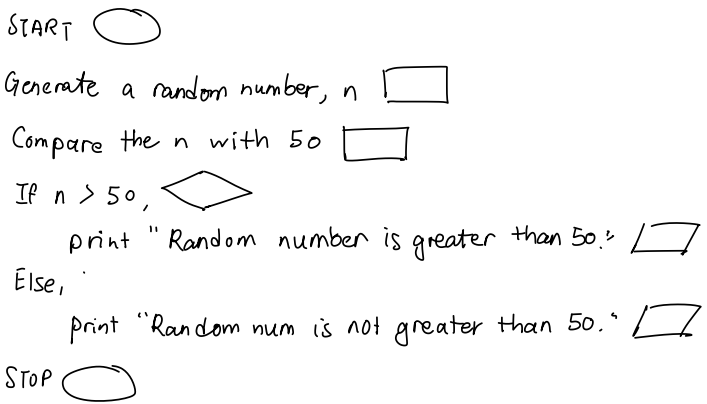
START
Enter num1, num2
Calculate product = num1 * num2
Display product
Stop

```
import java.util.Scanner;

public class Q1 {
    public static void main(String[]args){
        Scanner tekan = new Scanner(System.in);
        int num1 = tekan.nextInt;
        System.out.println("Enter num1 :");
        System.out.println("Enter num2 :");
        int num2 = tekan.nextInt;
        int product = num1 * num2;
        System.out.println(num1 +"x"+num2+"="+product);
    }
}
```

2

I : none
P : generate a random number, n
    compare n with 50
O : n is greater than 50
    n is not greater than 50




```
import java.util.Random;

public class Q2 {
    public static void main(String[] args) {
        Random cincai = new Random();
        int n = cincai.nextInt();
        If(n > 50);
            System.out.println("Greater.");
        Else;
            System.out.println("Not greater.");
    }
}
```

3

I : Read the mark, n
P : compare n with 40
O : If n >= 40, print "Pass"
    Else print "Fail"

START

Read the mark, n

If n >= 40'

print pass,

Else,

print "Fail"

STOP

```
import java.util.Scanner;
public class Q3{
    public static void main (String[] args){
        Scanner Tekan = new Scanner (System.in);
        int markah = tekan.nextInt();
        if (markah >=40) {
            System.out.println(" You pass!");
        } else {
            System.out.println("You fail!");
        }
    }
}
```

4

I : player1 score
player2 score

P : compare both score

O : player1 win, player2 win, draw

START

Read the score from p1

Read the score from p2

If p1 > p2,

Print "Player1 win"

Else if p2 > p1,

Print "Player2 win."

Else

Print "Draw".

STOP

```
import java.util.Scanner;

public class Q4 {
    public static void main (String[]args) {
        Scanner tekan = new Scanner(System.in);
        System.out.println("Player 1 score:");
        S1 = tekan.NextInt( );
        System.out.println("Player 2 score:");
        S2= tekan.NextInt( );
        If (S1 > S2) {
            System.out.println("Player1 win.");
        } else if (S2 > S1) {
            System.out.println("Player 2 win.");
        } else {
            System.out.println("Draw.");
        }
    }
}
```

5

I : length, width
P : perimeter = (2*length)+(2*width)
O : perimeter

START
Read length and width in centimeter
Calculate perimeter = (2*length)+(2*width)
Print perimeter
STOP

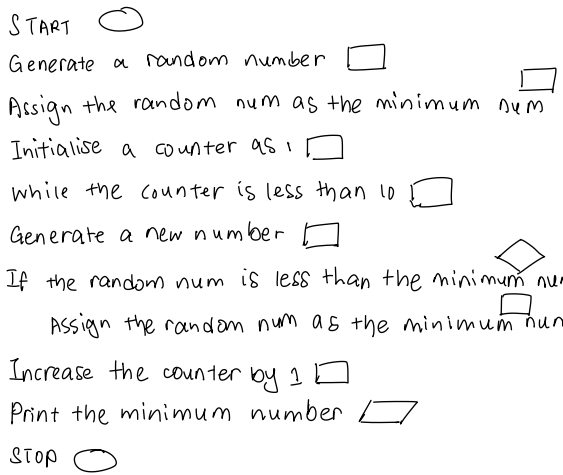
```
import java.util.Scanner;
public class Q5 {
    public static void main (String[]args) {
        Scanner tekan = new Scanner(System.in);
        System.out.println("Enter length:");
        int length = tekan.nextInt;
        System.out.println("Enter height:")
        int height = tekan.nextInt;
        int perimeter = (2*length)+(2*height);
        System.out.println("Perimeter is"+perimeter);
    }
}
```

6

I : none

P : ~~generate 10 random num~~
~~find the minimum num~~
Assign the first num as smallest num
Compare each new random number with the minimum num and update it

O : print the minimum num




```
import java.util.Random;

public class Q6 {

    public static void main (String[]args) {
    int nombor = tekan.nextInt();
    System.out.println(nombor);
    int smallest = nombor;
    for (int i=0 ; i<9 ; i++) {
        int again = tekan.nextInt();
        System.out.println(again);
        if (again < smallest) {
            smallest = again }
    }
    System.out.println("Smallest num is "+smallest)
}
```

7

I : none

P : generate 10 random number, n
if n %. 2 == 0, n is even num.
else, n is odd
count the numbers even and odd

O : Number of odd number
Number of even number

START

Initialise an oddNum = 0 and EvenNum = 0

Initialise counter as 1

while the count is less than or equal 10

Generate a random num, n within 10 - 100

If n%.2 == 0

increase evenNum to 1

Else
increase oddNum to 1

Increase counter by 1

Display number of evenNum and OddNum

STOP

8

I : A sentense

P : Read each char of the sentence
count the char of u and m

O : Number of U
Number of M

START

Get the sentence from user

Initialise alphabetU and alphabetM to 0

Initialise an index that points to the first char of the sentence

While the index is less than the length of sentence

- Read the chars according to the index
- If the char is U
  - Increase the alphabetU by 1
- If the
  - Increase the alphabetM by 1
- Increase index by 1

Print the num of alphabetU and alphabetM found

```
import java.util.Scanner;
public class Q8{
    public static void main(String[] args){
        Scanner tekan= new Scanner(System.in);
        int numU=0;
        int numM=0;
        System.out.println("Enter a sentence:");
        String sentence = tekan.nextLine();
        for(int index=0;index<sentence.length;index++){
            char check=sentence.charAt(index);
            if(check='u'||check=='U'){
                numU++;
            }else if(check='m'||check='M'){
                numM++;
            }
        }
        System.out.println("Num of U is"+numU);
        System.out.println("Num of M is"+numM);
    }
}
```

1). length

2). charAt(index)

q

I: webpage , keyword

P: Read the webpage line by line
compare each word with keyword
and count the match

O: Frequency of the keyword

START
Read the webpage
Read the keyword given
Initialise keywordCount = 0
Initialise the index point to the first word
While the index < total num of words
    Read the word according to the index
    If the word same as the keyword
        Increase the keywordCount by 1
    Increase the index by 1
Print the frequence of the keyword
STOP

```
import java.util.Scanner;
public class Qq {
    public static void main (String[] args) {
        Scanner Tekan = new Scanner (System.in);
        System.out.println ("Enter the webpage text:");
        String webpage = Tekan.nextLine();
        System.out.println ("Enter the keyword:")
        String webpage = Tekan.next();
        int keywordCount = 0;
        String [] words = webpage.split (" ");
        for (int index = 0; index < words.length; index ++) {
            String currentWord = words [index];
            If (currentWord.equalsIgnoreCase (keyword)) {
                keywordCount ++;
            }
        }
        System.out.println("Frequency of the keyword: "+ keywordCount);
    }
}
```

10

I : None

P : Generate 100 students and store them in a list
count the students whose gender is female

O : Number of female student

START
Initialise the counter as 0
Initialise femaleCount as 0
while the counter less than 100
Randomly generate a student with gender
Store the student in the list
Increase the counter by 1
If the student is female
FemaleCount = FemaleCount +1
Display FemaleCount
STOP

method A
"F" or "M"

method B
"0" or "1"

```
import java.util.Random
import java.util.ArrayList;
public class Q1 {
    public static void main (String[] args) {
        Random rnd = new Random()
        自訂的，不是 VarName 也不是 reserved word
        ArrayList<Character> students = new
            ArrayList<>();
        int FemaleCount = 0;
        for(int i=0; i<100; i++){
            char gender = rnd.nextBoolean()?'F':'M';
            students.add(gender);
        }
        for(int i=0; i< students.size(); i++) {
            if (students.get(i) == 'F') {
                FemaleCount ++;
            }
        }
        System.out.println("Num of female are:" + FemaleCount);
    }
}
```

11

I : None

P: Generate 5 random num and store them in a list
Store the list in descending order (bubble sort)

O: The list of number in descending order

START
Generate 5 random integer into numbers
Sort numbers in descending order
Initialise index = 0
while index < 5 ,
Display number [index]
index = index - 1
END while
STOP

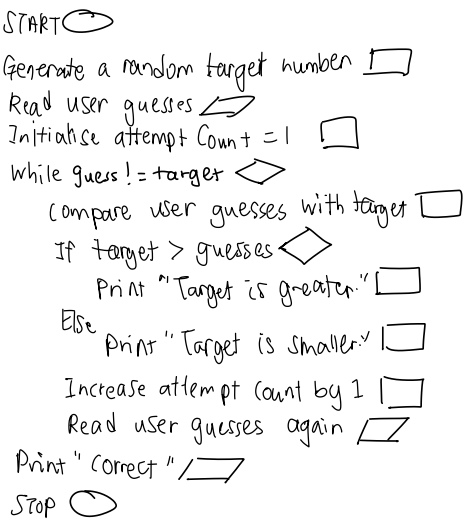
```
import java.util.*;
public class Q11 {
    public static void main (String[] args) {
        Scanner tekan = new Scanner (System.in);
        Random cincai = new Random();
        int [] numbers = new int [5];
        for(int i=0; i<5; i++) {
            numbers[i]=cincai.nextInt(0,101);
        }
        System.out.println("Original list are "+
                Arrays.toString(numbers));
        Arrays.sort(numbers);
        for (int i=5; i>0; i--) {
            System.out.println(numbers[i]+" ");
        }
    }
}
```

12

I: user guesses

P: Randomly generate a fix target number
Initialise attempt Count = 0
Compare user guesses with target
Repeat while unequal

O: Higher, Lower, Correct, attempt count




```
import java.util.*;
public class Q12 {
    public static void main(String[]args){
        Scanner tekan = new Scanner (System.in);
        Random cincai = new Random();
        int target = cincai(0,101);
        System.out.println("Let guess:");
        int guess = tekan.nextInt();
        int attemptCount =1;
        while(guess != target) {
            if (target>guess) {
                System.out.println("Target is Greater.")
            }else{
                System.out.println("target is smaller.");
            }
            attemptCount ++;
            System.out.print("Guess again:");
            guess = tekan.nextInt();
        }
        System.out.println("Correct, your attemp
            count is" + attemptCount);
    }
}
```